MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**B.Tech**	**Course Type**	**OE**
Course Name	**Air and Noise pollution**	**Course Code**	**CIE 4301**
Semester	**VII**	**Max Marks**	**50**
Date	**02/12/2024**	**Duration**	**3hr**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Discuss about the Bhopal gas tragedy and its causative agents.	**3**	1	**1,6,7,8,12**	**2**
1B	Define smog. Explain the different types of smog.	**3**	1	**1,6,7,8,12**	**2**
1C	Distinguish between stationary and mobile source of air pollutants.	**4**	1	**1,6,7,8,12**	**4**
2A	Enlist the factors responsible for scavenging the air pollutants from the atmosphere.	**2**	2	**1,6,7,8,12**	**2**
2B	Define meteorology. Explain the importance of meteorology in regional air quality management.	**4**	2	**1,6,7,8,12**	**4**
2C	Define wind rose. Explain the significance of wind speed and direction in air pollution dispersion.	**4**	2	**1,6,7,8,12**	**4**
3A	Explain any four factors to be considered for industrial plant location.	**4**	3	**1,6,7,8,12**	**4**
3B	With a neat sketch explain the working of high volume air sampler.	**3**	3	**1,6,7,8,12**	**4**
3C	Explain types of air pollution sampling based on source of air pollutant.	**3**	3	**1,6,7,8,12**	**4**
4A	Describe the evolution of vehicular emission norms in India.	**4**	4	**1,6,7,8,12**	**2**

<table>
<tr><td>4B</td><td>Explain the role of nitrogen dioxide and ultraviolet light in the formation of photochemical smog with equations.</td><td>3</td><td>4</td><td>1,6,7,8,12</td><td>4</td></tr>
<tr><td>4C</td><td>What causes acid rain? Explain the effects of acid rain on materials and the ecosystem.</td><td>3</td><td>4</td><td>1,6,7,8,12</td><td>4</td></tr>
<tr><td>5A</td><td>Explain if asbestos can be considered as an air pollutant. What are the health risks associated with fluoride poisoning?</td><td>4</td><td>4</td><td>1,6,7,8,12</td><td>4</td></tr>
<tr><td>5B</td><td>The table below shows the concentration of two pollutants and the exposure duration. The Time limiting value (TLV) value of Pollutant A is 20 ppm, and Pollutant B is 4.5 ppm. Determine which pollutant is harmful to the workers for an 8-hour Time weighted average (TWA) period.
<table>
<tr><th>Duration of exposure (hr)</th><th>Measured Concentration of Pollutant A(ppm)</th><th>Measured Concentration of Pollutant B(ppm)</th></tr>
<tr><td>3</td><td>10</td><td>6</td></tr>
<tr><td>2</td><td>4</td><td>4</td></tr>
<tr><td>3</td><td>5</td><td>4</td></tr>
</table></td><td>2</td><td>4</td><td>1,6,7,8,12</td><td>3</td></tr>
<tr><td>5C</td><td>Discuss on the health effects of noise pollution. Suggest noise-controlling barriers that can be adopted to reduce noise pollution.</td><td>4</td><td>5</td><td>1,6,7,8,12</td><td>4</td></tr>
</table>

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***10***
2	***10***
3	***10***
4	***16***
5	***4***

BL	Total Marks
2	**12**
3	**2**
4	**36**
5	**0**